

Function notation



- 1 The point $(7, -6)$ satisfies the function $f(x)$. Write this using function notation.
- 2 If $f(x) = 9x^2 + 7x - 4$, find:
 - a $f(-4)$
 - b $f(10)$
- 3 If $f(x) = -6x + 4$, find:
 - a $f(4)$
 - b $f(0)$
- 4 If $f(x) = 4x + 4$, find:
 - a $f(2)$
 - b $f(-5)$
- 5 If $f(x) = 3x - 1$, find:
 - a $f(3)$
 - b $f(-4)$
- 6 If $g(x) = \frac{7x}{4}$, find:
 - a $g(5)$
 - b $g(-4)$
- 7 Consider the function $f(x) = 4 + x^3$.
 - a Evaluate $f(4)$
 - b Evaluate $f(-2)$
- 8 Consider the function $f(x) = 2x^2 - 2x + 5$. Evaluate $f\left(\frac{1}{2}\right)$.
- 9 Consider the function $f(x) = \sqrt{5x + 9}$. Find the exact value of:
 - a $f(0)$
 - b $f(2)$
 - c $f(-1)$
- 10 If $f(t) = \frac{t^3 + 27}{t^2 + 9}$, evaluate the following:
 - a $f(-3)$
 - b $f(3)$
 - c $f(4)$
- 11 Consider the function $f(x) = x^2 + 8x$. Write an expression for:



12 Consider the function $f(x) = 2x^3 + 3x^2 - 4$.

a Evaluate $f(0)$.

b Evaluate $f\left(\frac{1}{4}\right)$.

13 If $j(x) = 3^x - 3^{-x}$, find the following, rounding your answers to two decimal places if necessary:

a $j(0)$

b $j(1)$

c $j(4)$

14 If $m(x) = \sqrt{12^2 - x^2}$, find:

a $m(4)$

b $m(0)$

c $m(9)$

d $m(\sqrt{2})$

 **15** Consider the function $p(x) = x^2 + 8$.

a Evaluate $p(2)$.

b Form an expression for $p(m)$.

 **16** Consider the equation $x + 3y = 6$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f(x)$ for y .

c Find the value of $f(3)$.

 **17** Consider the equation $x - 4y = 8$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f : x \rightarrow$.

c Find the value of $f : 12$.

18 Consider the equation $y + 6x^2 = 3 - x$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f(x)$.

c Find the value of $f(3)$.

19 Consider the equation $y - 4x^2 = 5 + x$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f : x \rightarrow$.

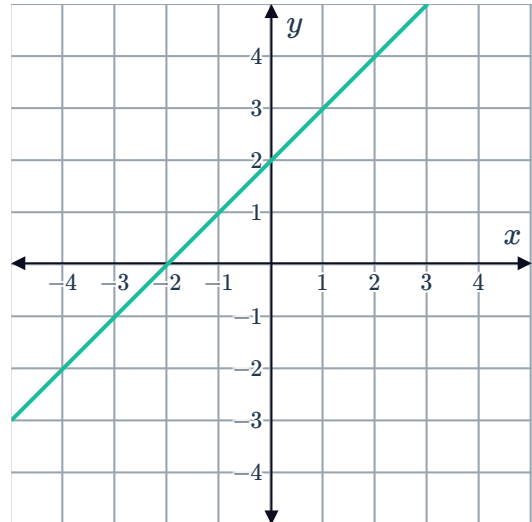


20 Consider the equation $-6x + 5y = 7$.

- a Make y the subject of the equation.
- b Rewrite the equation using function notation $f : x \rightarrow$.
- c Find the value of $f : 3$.

21 Use the graph of the function $f(x)$ to find each of the following values:

- a $f(0)$
- b $f(-2)$
- c The value of x such that $f(x) = 3$



22 Consider the relation $h : x \rightarrow -x^2 + 6x - 6$.

- a For any input value of x , state the maximum number of distinct output values h can produce.
- b Is h a function? Explain your answer.

23 For each of the following pairs of variables, determine which is the independent variable and which is the dependent variable:

- a Cost of pizza and size of pizza.
- b Mark achieved on a test and time spent studying.
- c Duration of a loan and amount borrowed.
- d Time spent exercising and physical fitness level.

24 Consider the function $f : x \rightarrow x^2 - 49$.

- a Find:
 - i $f(1)$
 - ii $f(8)$
 - iii $f(0)$
- b If $f : x \rightarrow 12$, what are the possible values for x , rounded to two decimal places?

25 Consider the function $g(x) = ax^3 - 3x + 8$



- a** Form an expression for $g(k)$.
- b** Form an expression for $g(-k)$.
- c** Is $g(k) = g(-k)$?
- d** Is $g(k) = -g(-k)$?

Applications

 **26** If $Z(y) = y^2 + 12y + 32$, find y when $Z(y) = -3$.

27 A function f is defined by $f : x \rightarrow (x + 4)(x^2 - 4)$.

- a** Evaluate $f(6)$.
- b** Find all solutions for which $f : x \rightarrow 0$.

28 A graph of a quadratic equation of the form $y = ax^2 + bx + c$ passes through the points $(0, -7)$, $(-1, -8)$ and $(4, 77)$.

- a** Using the point $(0, -7)$, find the value of c .
- b** Substitute c and $(-1, -8)$ into the equation $y = ax^2 + bx + c$ to obtain an equation that describes a in terms of b .
- c** Substitute c and $(4, 77)$ into the equation $y = ax^2 + bx + c$ to obtain a simplified second equation in terms of a and b .
- d** Solve for a and b .

29 The financial team at The Gamgee Cooperative wants to calculate the profit, $P(x)$, generated by producing x units of wetsuits.

The revenue produced by the product is given by the equation $R(x) = -\frac{x^2}{4} + 40x$. The cost of production is given by the equation $C(x) = 5x + 410$.

The profit is calculated as $P(x) = R(x) - C(x)$.

- a** Find an expression for $P(x)$ in terms of x .
- b** Find the values of the following:
 - i** $R(70)$
 - ii** $C(70)$
 - iii** $P(70)$
- c** Sketch the graphs of $y = R(x)$, $y = C(x)$ and $y = P(x)$.

- 30** Elizabeth wants to calculate the cost to travel to Tehran and then Mexico City at certain times of the year. A program on her computer calculates $T(x)$, the total cost of this trip, by adding the cost of travelling to Tehran, $S(x)$, and the cost of travelling to Mexico City, $U(x)$.

Based on historical data, the computer program uses the calculations:

$$S(x) = x^2 - 200x + 10\,227 \quad \text{and} \quad U(x) = 18x + 15\,423$$

where x is the number of days until the date of travel.

- a** Find an equation for $T(x)$.
- b** Find the values of the following:
 - i** $S(91)$
 - ii** $U(91)$
 - iii** $T(91)$
- c** Sketch the graphs of $y = S(x)$, $y = U(x)$ and $y = T(x)$.